

CHRISNERG MQOBO

Software Engineer

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github.com/Chrisnerg | linkedin.com/in/chrisnerg-mqobo | [Portfolio](#)

PROFESSIONAL SUMMARY

Software Engineer with 1 year of professional experience building, shipping, and supporting production software across cloud, web, desktop, and real-time environments. Strong in TypeScript, React, Next.js, Node.js, SQL Server (T-SQL), Electron, REST APIs, WebSockets, and Playwright. Currently contributing to the continuous development and weekly rollout of an enterprise cloud platform, working across frontend, backend, database, testing, deployment, production support, and developer onboarding.

EXPERIENCE

Junior Software Engineer — Tontrac Systems Analytics

September 2025 – Present

Sandton, Gauteng

- Contribute to the ongoing development of a next-generation enterprise cloud platform, delivering production features in a weekly release cycle using Next.js, React, TypeScript, TanStack Query, Node.js services, and SQL Server.
- Deliver end-to-end features spanning user interfaces, API/service integrations, validation, permission controls, server-state management, and SQL Server stored-procedure-backed workflows.
- Develop and maintain real-time monitoring and desktop systems using React, TypeScript, Electron, Node.js, secure IPC, and WebSockets across 50+ operational sites.
- Build Playwright end-to-end test suites for critical production workflows, reducing repetitive manual regression testing and improving release confidence.
- Contribute to Python/OpenCV computer-vision functionality, including OCR and license-plate recognition for automated weighbridge data capture and vehicle validation.
- Engineer Windows OS integrations using Registry APIs and Win32 messaging to automate system configuration and reduce repetitive client-machine setup.
- Perform production deployments, configuration, troubleshooting, and technical support while collaborating with engineering, IT Support, and Control Room teams.
- Support onboarding and technical development of a 12-month YES Programme intern, providing guidance on the tech stack, internal systems, development workflows, debugging, and business operations alongside senior engineers.

Informatics Tutor — University of Johannesburg

February 2025 – September 2025

Auckland Park, Johannesburg

- Mentored 100+ undergraduate students in programming fundamentals, debugging, and algorithmic thinking, collaborating with academic staff on curriculum workshops.

PROJECTS

URL Shortener — Full Stack Web Application

github.com/Chrisnerg/url-shortener-fullstack

- Built a full stack URL shortening service with click analytics using TypeScript, Node.js, Express.js, and React; designed REST APIs with input validation, short-code generation, and persistent storage.

Web Content Scraper — Automation Tool

github.com/Chrisnerg/Web-Content-Scraper

- Developed a Node.js tool that automates extraction of structured content from web pages.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, SQL (T-SQL), Python, HTML5, CSS3

Frontend: React, Next.js, Tailwind CSS, TanStack Query, Responsive UI

Backend: Node.js, Express.js, REST APIs, WebSockets, Authentication, Data Validation

Databases: SQL Server, PostgreSQL

Desktop & Systems: Electron, IPC Architecture, Real-Time Systems, Windows Registry / Win32 APIs

Computer Vision: OpenCV, OCR, License-Plate Recognition

Testing & DevOps: Playwright, End-to-End Testing, Git, GitHub, Docker, CI/CD, Postman

EDUCATION & CERTIFICATIONS

BSc Computer Science and Informatics — University of Johannesburg

Completed May 2026

Minor in Mathematics and IT Management | Auckland Park, Johannesburg

AWS Certified Developer – Associate — In Progress, 2026